2 Additional Simulation Results - Toeplitz

In the following subsections additional results to Puchhammer et al. (2025) are presented. The methods are analyzed in five main scenarios:

1. $N = 2$ balanced groups (our basic scenario with $p = 10, n_1 = n_2 = 100$),
2. $N = 5$ balanced groups ($p = 10, n_1 = \ldots, n_5 = 100$),
3. $N = 2$ unbalanced groups ($p = 10, n_1 = 100, n_2 = 50$),
4. $N = 2$ balanced groups with increasing singularity issues ($p = 20, n_1 = n_2 = 30$) and
5. high-dimensional $N = 2$ balanced groups ($p = 60, n_1 = n_2 = 40$).

The covariance structure analyzed here is of Toeplitz type (similar to Raymaekers and Rousseeuw, 2023), and we take $\boldsymbol{\Sigma}_{k,ij} = \zeta_k^{|i-j|}$ $(i, j = 1, \ldots, p)$ where ζ_k is randomly drawn from a uniform distribution in $[0.5, 1]$.

For each setting, the performance of parameter estimation compared to competing methods is visualized as well as the correctness of flagging outlying cells. Moreover, for each scenario a table with the number of repetitions considered in the figures for cellMCD and cellGMM is given. They can deviate from the default number of 100 due to the restriction of the cellMCD regarding the number of marginal outliers and computational issues for the cellGMM in higher dimensional data ($p = 40, 60$).

2.1 Scenario 1 - Basic Balanced Scenario

γ_{cell}	$\mu = 0$		$\mu =$ varying	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	100	100	84	100
6	100	100	61	99
10	100	100	58	98

Table 1: Number of successful replications for cellMCD in scenario 1 for Toeplitz covariance structure.

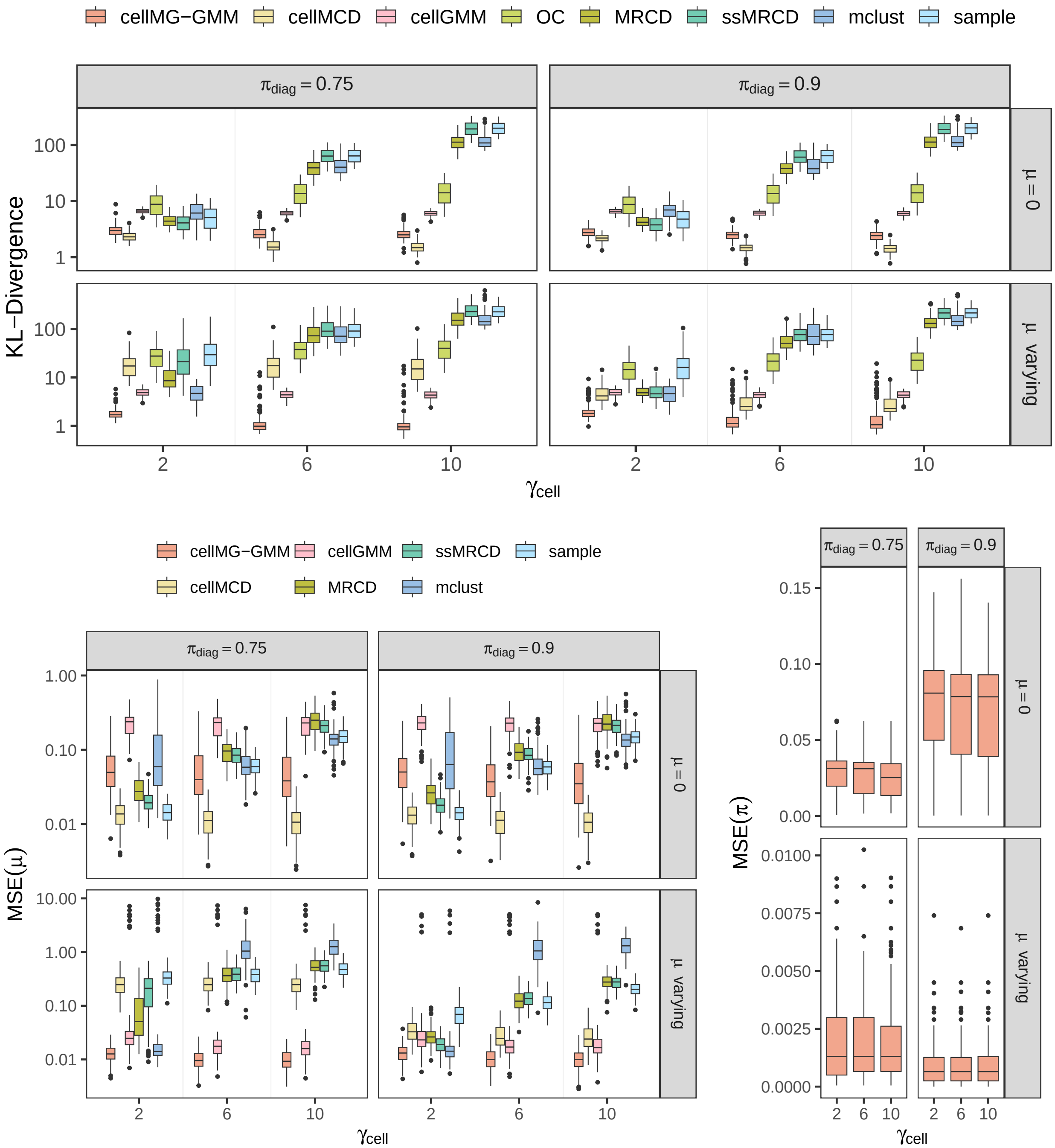

Figure 1: Parameter estimates for scenario 1 ($N = 2, p = 10, n_1 = n_2 = 100$) with Toeplitz structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities $\boldsymbol{\pi}$.

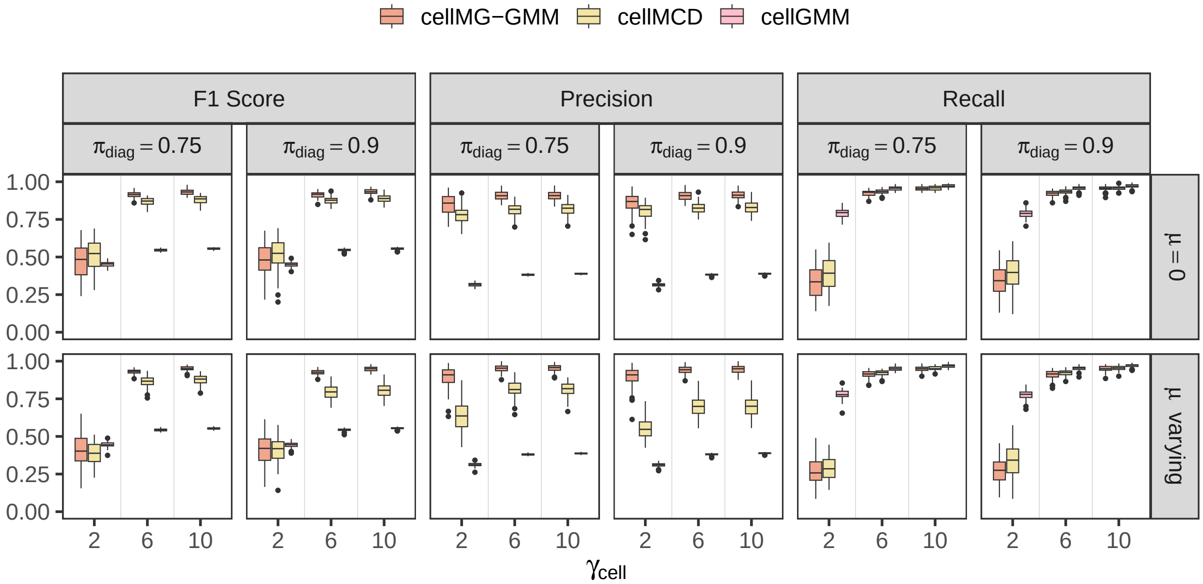

Figure 2: Performance of cellwise outlier detection in scenario 1 ($N = 2, p = 10, n_1 = n_2 = 100$) with Toeplitz structured covariances evaluated by precision, recall and F1-score.

2.2 Scenario 2 - Balanced Scenario with 5 groups

γ_{cell}	$\mu = 0$		$\mu =$ varying	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	100	100	61	100
6	100	100	21	100
10	100	100	16	99

Table 2: Number of successful replications for cellMCD in scenario 2 for Toeplitz covariance structure.

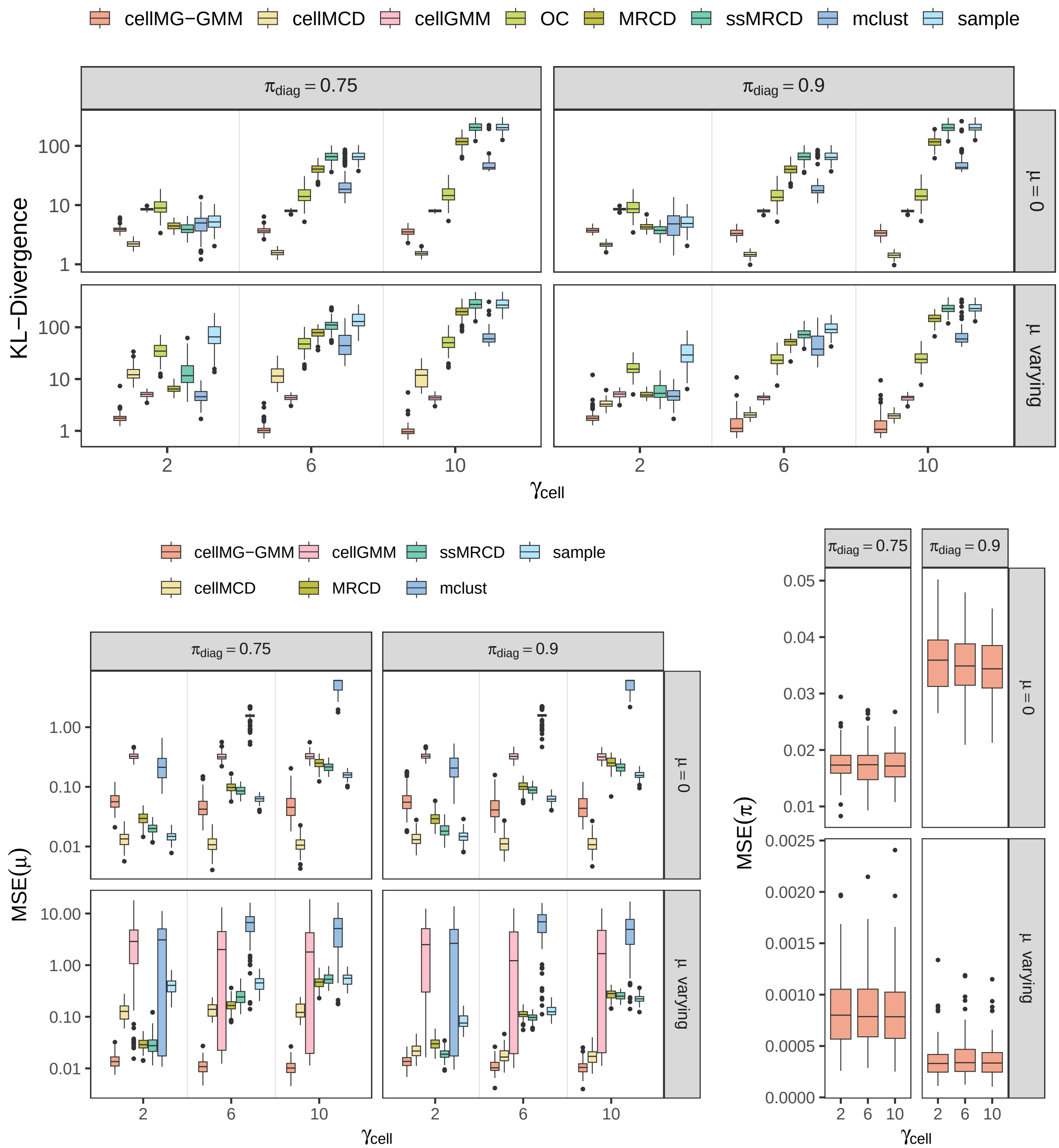

Figure 3: Parameter estimates for scenario 2 ($N = 5, p = 10, n_1 = n_2 = 100$) with Toeplitz structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities $\boldsymbol{\pi}$.

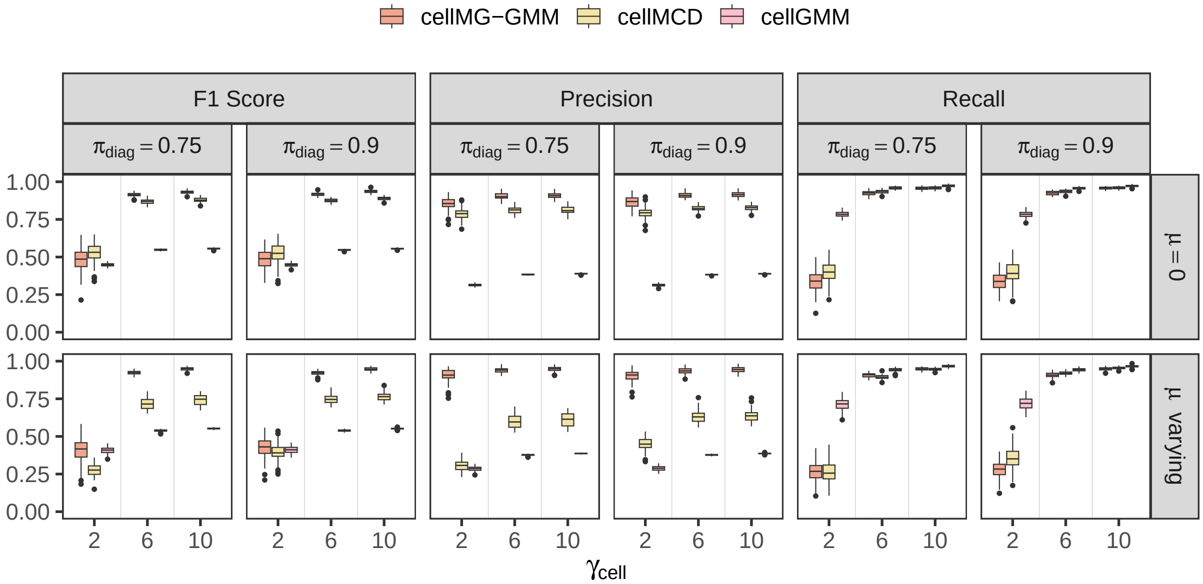

Figure 4: Performance of cellwise outlier detection in scenario 2 ($N = 5, p = 10, n_1 = n_2 = 100$) with Toeplitz structured covariances evaluated by precision, recall and F1-score.

2.3 Scenario 3 - Unbalanced Scenario

γ_{cell}	$\mu = 0$		$\mu =$ varying	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	100	100	84	100
6	100	100	68	96
10	100	100	58	93

Table 3: Number of successful replications for cellMCD in scenario 3 for Toeplitz covariance structure.

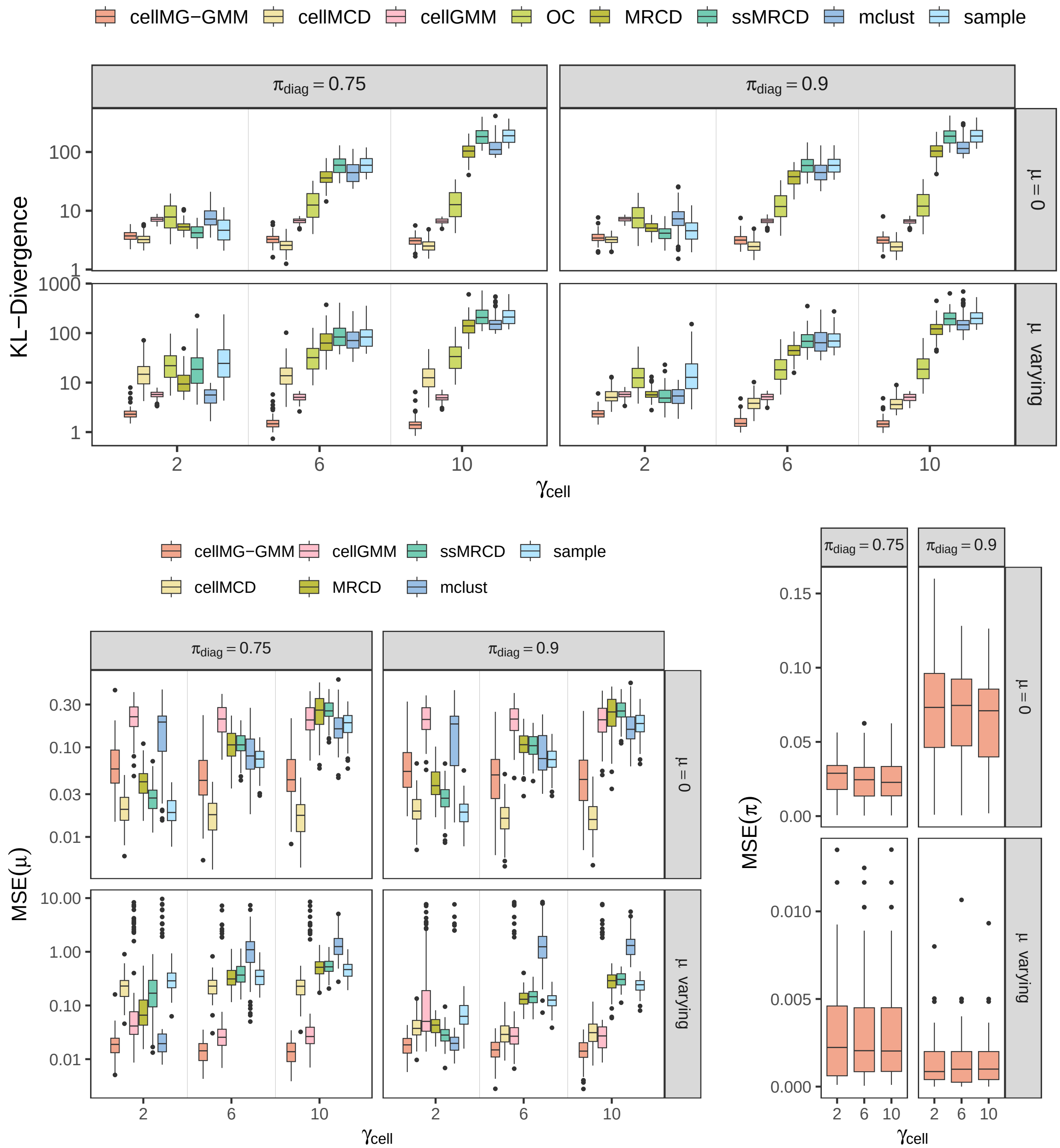

Figure 5: Parameter estimates for scenario 2 ($N = 5, p = 10, n_1 = n_2 = 100$) with Toeplitz structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities $\boldsymbol{\pi}$.

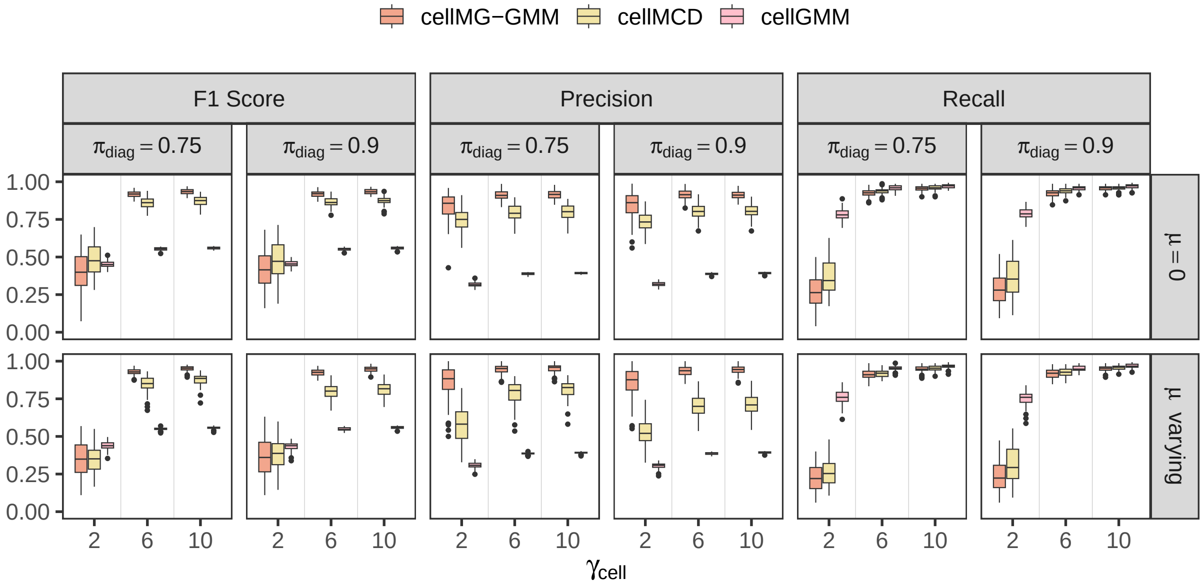

Figure 6: Performance of cellwise outlier detection in scenario 2 ($N = 5, p = 10, n_1 = n_2 = 100$) with Toeplitz structured covariances evaluated by precision, recall and F1-score.

2.4 Scenario 4 - Increasing Singularity Issues

γ_{cell}	$\mu = 0$		$\mu =$ varying	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	89	88	38	85
6	84	85	14	55
10	84	84	12	53

Table 4: Number of successful replications for cellMCD in scenario 4 for Toeplitz covariance structure.

γ_{cell}	$\mu = 0$		$\mu =$ varying	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	52	54	24	23
6	44	44	30	36
10	45	49	36	35

Table 5: Number of successful replications for cellGMM in scenario 4 for Toeplitz covariance structure.

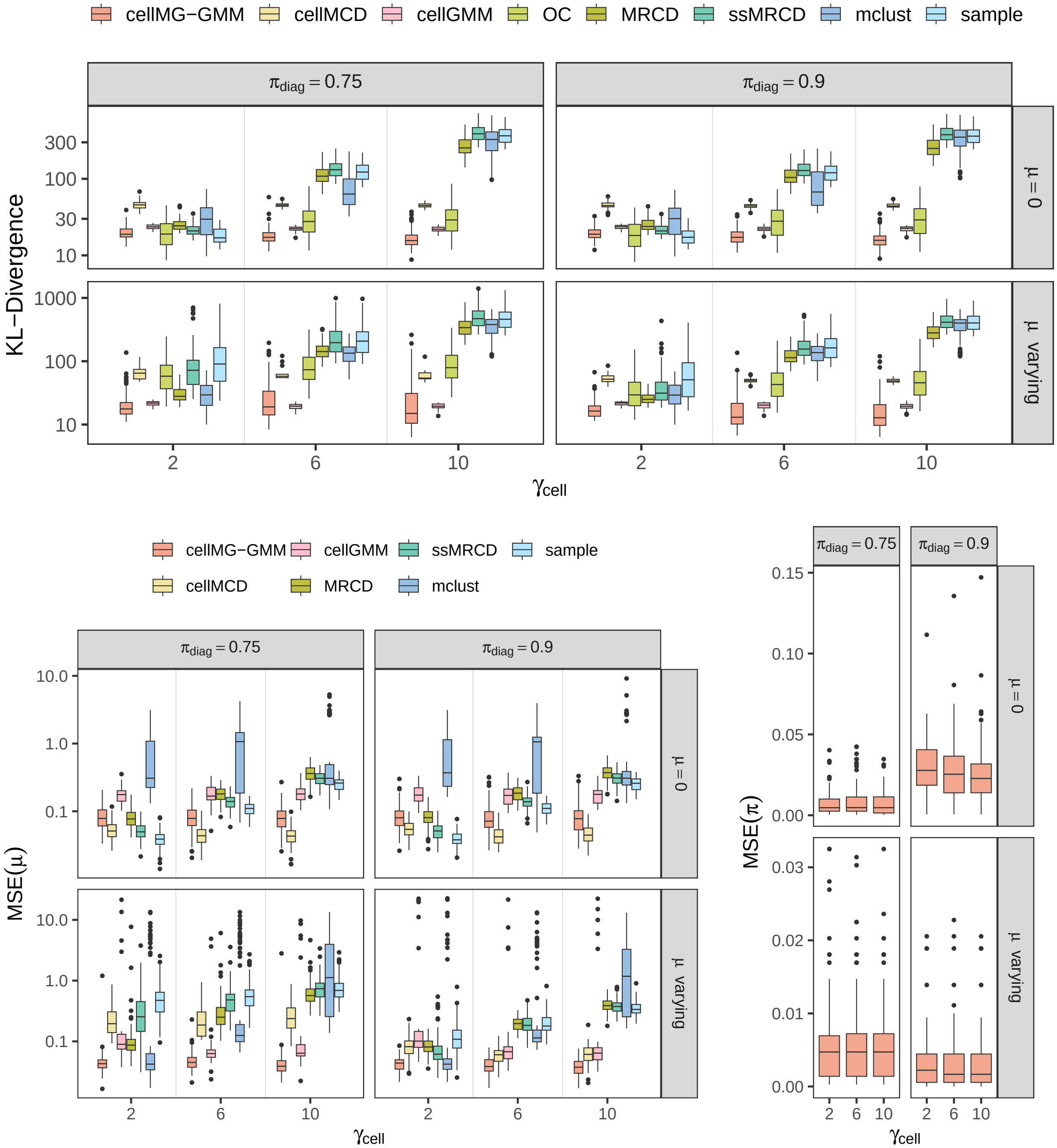

Figure 7: Parameter estimates for scenario 4 ($N = 2, p = 10, n_1 = 100, n_2 = 50$) with Toeplitz structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities $\boldsymbol{\pi}$.

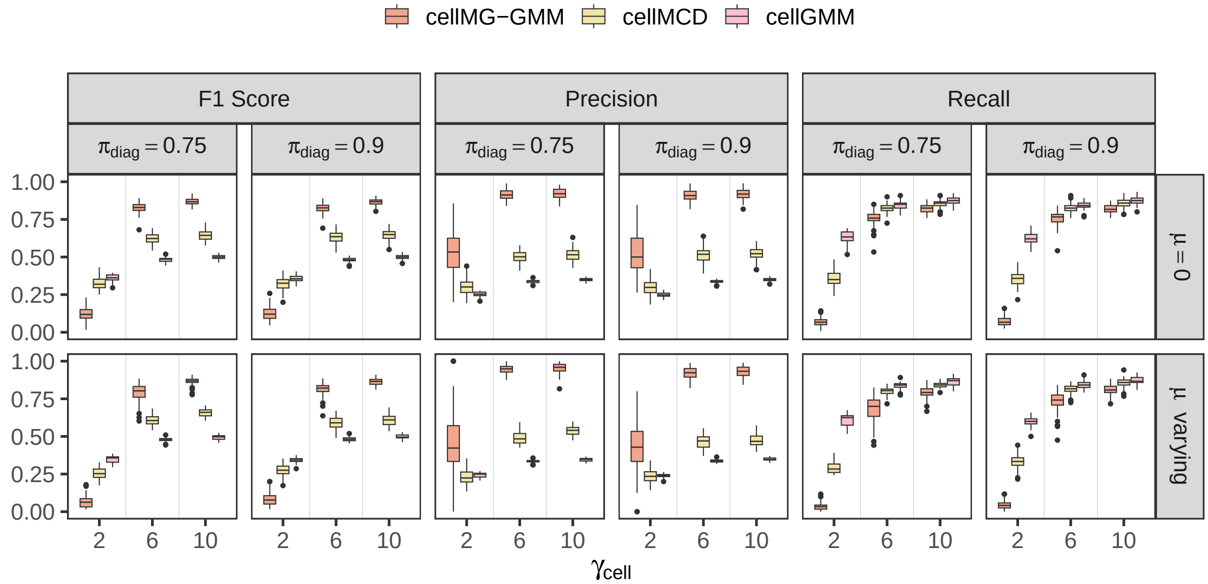

Figure 8: Performance of cellwise outlier detection in scenario 4 ($N = 2, p = 10, n_1 = 100, n_2 = 50$) with Toeplitz structured covariances evaluated by precision, recall and F1-score.

2.5 Scenario 5 - High-Dimensional

No calculations of cellMCD and cellGMM were successful and are thus not included.

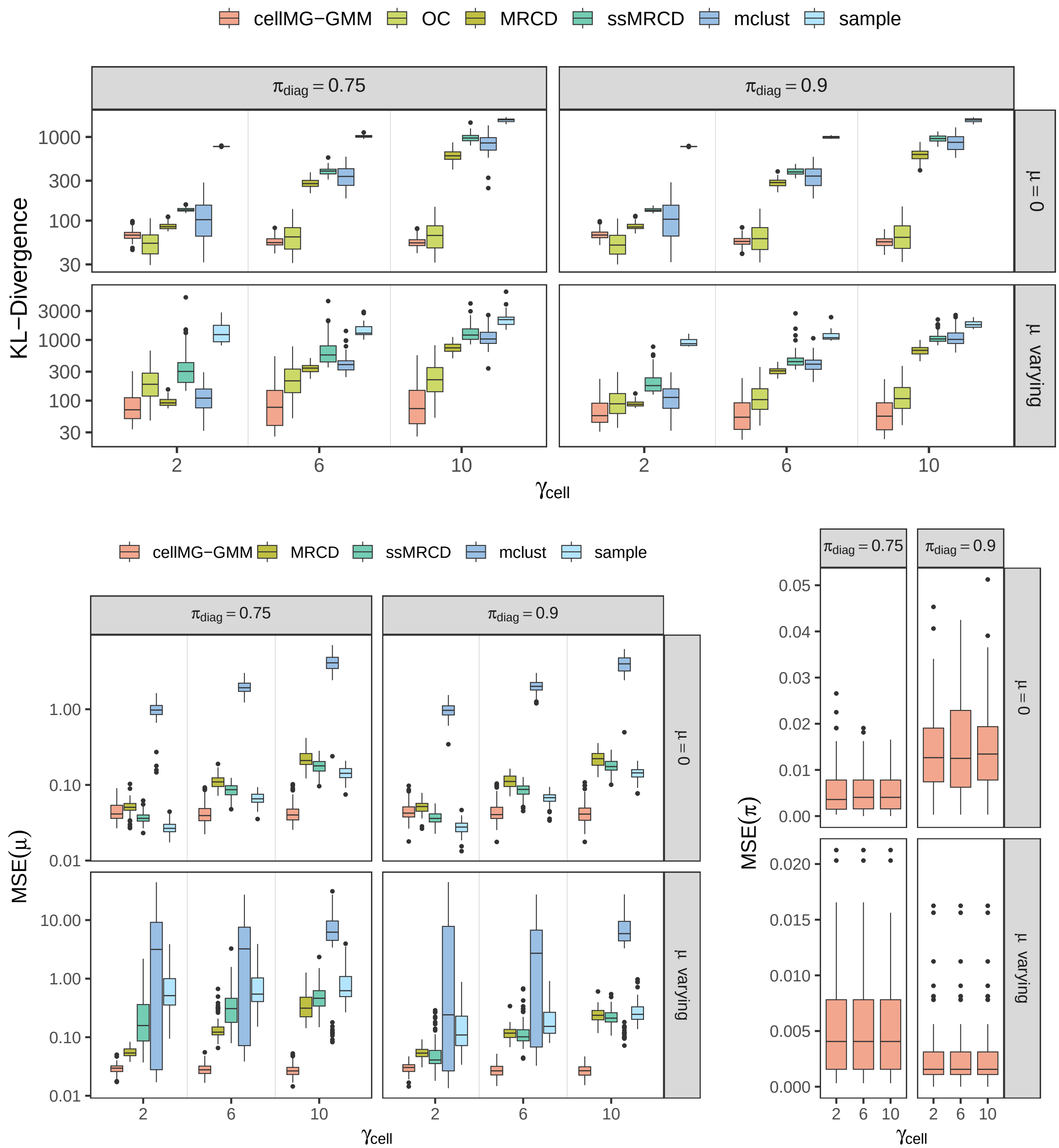

Figure 9: Parameter estimates for scenario 5 ($N = 2, p = 60, n_1 = n_2 = 40$) with Toeplitz structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities $\boldsymbol{\pi}$.

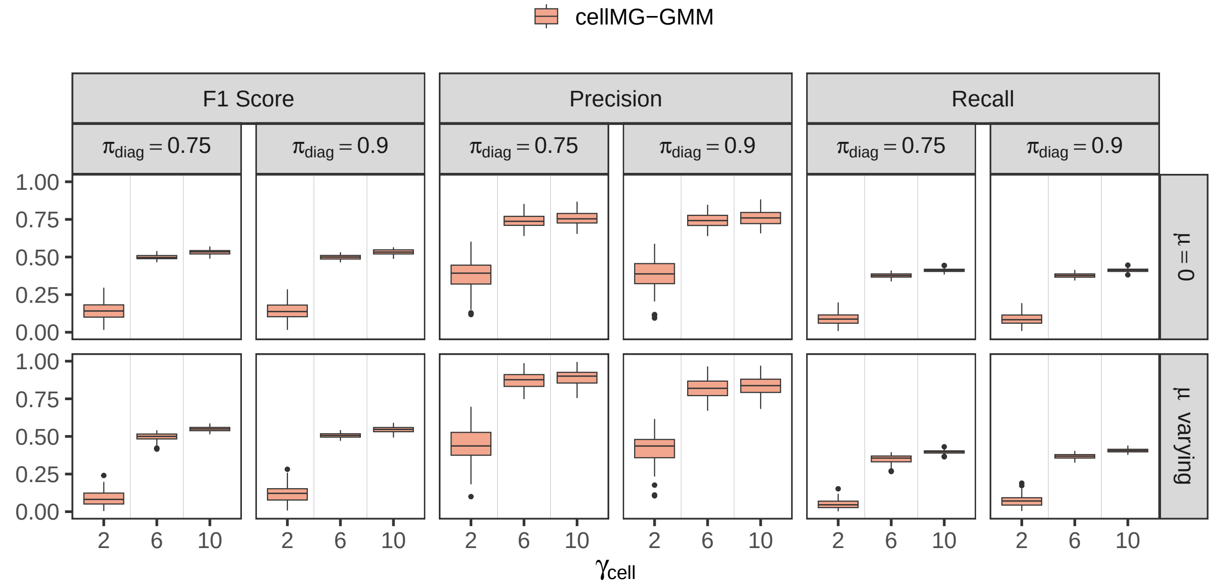

Figure 10: Performance of cellwise outlier detection in scenario 5 ($N = 2, p = 60, n_1 = n_2 = 40$) with Toeplitz structured covariances evaluated by precision, recall and F1-score.

References

Puchhammer, P., Wilms, I., and Filzmoser, P. (2025). Outlier-robust multi-group gaussian mixture modeling with flexible group reassignment. arXiv preprint arXiv:2504.02547.

Raymaekers, J. and Rousseeuw, P. J. (2023). The cellwise minimum covariance determinant estimator. Journal of the American Statistical Association, pages 1–12.